Subway Opening in the Process of Urban Revival

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Replication package: `master.py` in the repository root reproduces Figure 1.

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Setting: Hudson Yards in New York City was once underdeveloped compared to other Manhattan neighborhoods, primarily consisting of unused land and rail yards owned by the MTA. Development began in 2012, becoming one of the largest privately-led real estate projects (Fisher and Leite, 2018). In September 2015, the 34th Street–Hudson Yards subway station opened. As the last case of a new station built in a completely new neighborhood in NYC, it offers a unique case study of how commercial development interacts with urban infrastructure. This study examines how the opening of the subway influences commuter patterns in the area.

Data: To estimate changes in commuting patterns, I use two datasets: the Primary Land Use Tax Lot Output (PLUTO) from the New York City Department of City Planning (2025) and the LEHD Origin-Destination Employment Statistics (LODES) from the U.S. Census Bureau (2025). PLUTO contains data on all tax lots in New York City, updated annually to reflect demolitions and new constructions. I aggregate the data to the census tract level for the Hudson Yards area to track neighborhood development. LODES provides a cross-product of census blocks in New York State. I aggregate this data to the tract level and identify workers commuting to Hudson Yards from each New York census tract. I then estimate the distance between tracts.

Fact: I observe two key trends in the data. First, the subway opening stimulates construction. Panel A of Figure 1 shows that after 2015, development in Hudson Yards accelerated. Second, it increases commutes for people living farther from the area, replacing those who live nearby. Panel B shows that, for all distance bins except the closest one, the number of commuters rose.

Discussion: The second fact is of main interest because it shows gentrification of the community in action. The newly built subway increases construction in the neighborhood and attracts new commuters because of more available transit and more opportunities.

Literature: The fact relates to previous literature that studies the effects of urban revival and development of urban infrastructure. Previous literature considered how rail transit development introduced advanced degree holders to the neighborhood (Liang et al., 2022). The other publications analyzed changes in housing prices (Guan et al., 2024) and ridership increase (Gonzalez-Navarro and Turner, 2018). My analysis is novel because it shows quantitatively how more people living far away from the neighborhood decide to commute because of new infrastructure and available transit. Previously, changes in commuters were not studied over a geographic distance.

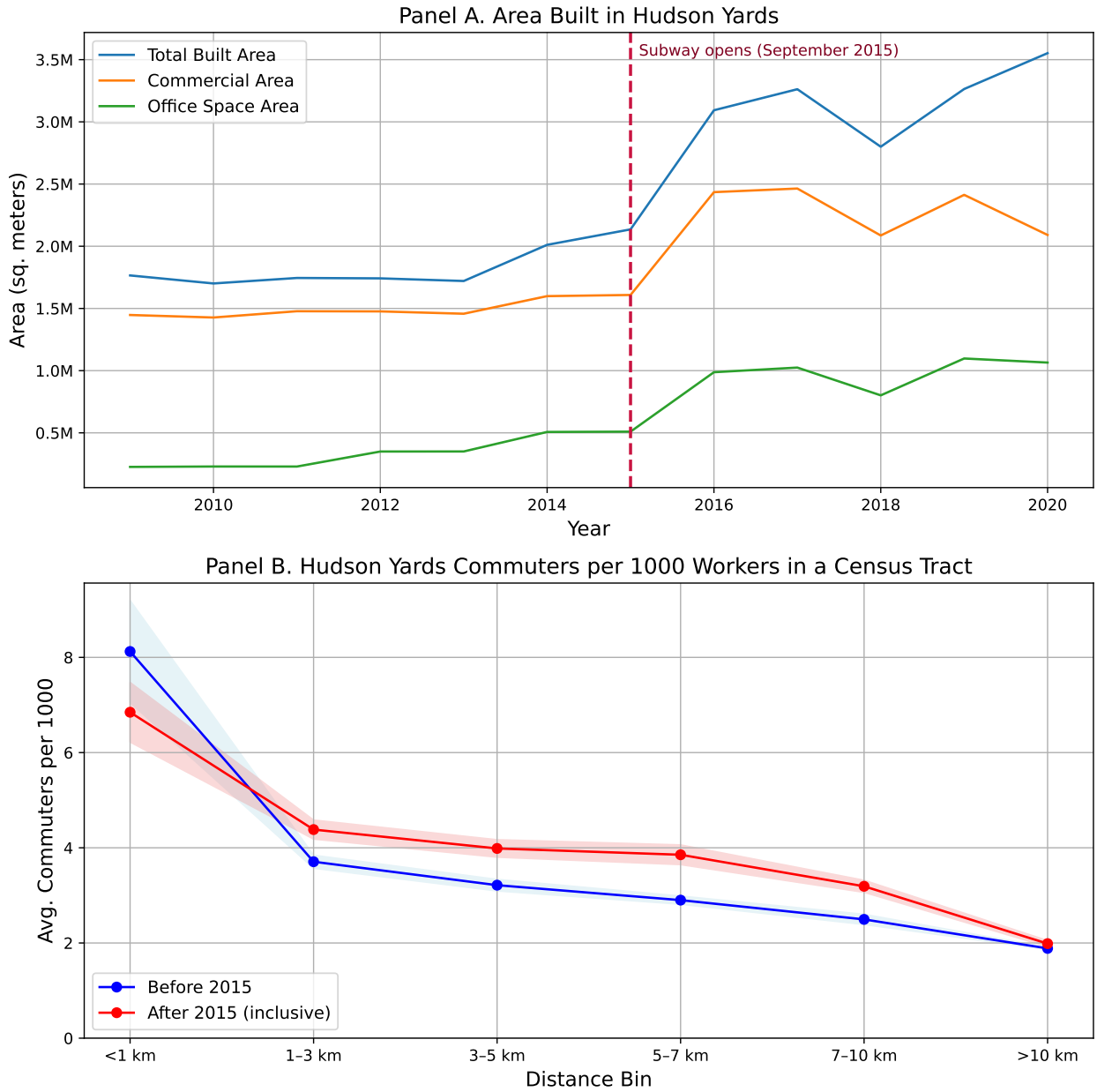


Figure 1: Impact of Subway on Construction and Commuting Patterns. Panel A shows changes in construction in Hudson Yards from 2009 to 2020, with the built area divided into three categories: total area, commercial space, and office space. The areas are converted from square feet to square meters, and the vertical line marks the subway opening. Panel B illustrates changes in commuting patterns to Hudson Yards before and after the subway opened. The y-axis represents the average number of Hudson Yards commuters per 1,000 workers in their census tract, while the x-axis bins census tracts based on their distance from Hudson Yards.

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